

04-10-2025

## Deferred Acceptance:

set of workers:  $w = \{w_1, \dots, w_N\}$

set of firms:  $f = \{f_1, \dots, f_N\}$

Assume: strict preferences

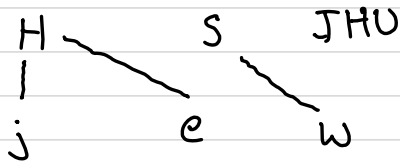
A matching  $\mu: W \cup F \longrightarrow W \cup F$   
- 1 to 1 match  $w \rightarrow f; f \rightarrow w$

## Stability:

- no worker firm pair preferring each other to match
- no worker prepares to be unmatched relative to current match
  - ↳ same for firm

## Algorithm:

- $w_i$  propose to favored  $f_i$
- $f_i$  then hold
  - if 1 proposal, accept or be unmatched
  - if  $> 1$ , accept favored, reject rest.
- re-run those who reject
- $f_i$  then re-evaluate:
  - hold favored, reject rest
- run until all matched



$j: H \succ S \succ J$   
 $e: H \succ S \succ J$   
 $w: S \succ H \succ J$

$H: j \succ w \succ e$   
 $S: e \succ w \succ j$   
 $J: w \succ j \succ e$

1:  $H \rightarrow j; S \rightarrow w; \emptyset \rightarrow e$

2:  $S \rightarrow e; H \rightarrow j; \emptyset \rightarrow w$

3:  $S \rightarrow e; H \rightarrow j; \emptyset \rightarrow w$

4:  $S \rightarrow e; H \rightarrow j; J \rightarrow w$

Algorithm always stable, but matching can change depending on who does proposing

Doctor optimal: no other stable match which Pareto improves doc. outcomes

Hosp. Optimal: same but hosp.

Pessimal: worst stable match for other side

- doc. opt. = hosp. pessimal

Optimal is for the side proposing

Always stable w.r.t provided preferences

# Proof by Contradiction:

Suppose the matching  $\mu$  generated by deferred acceptance is unstable s.t.  $\exists w_i, f_i$  with  $w_i \succ_i \mu(f_i)$   $f_i \succ_i \mu(w_i)$ . This occurs when  $w_i$  preferred  $f_i$  but was rejected for  $w_j$ , meaning there was not a feasible match above the current.



Goal: incentivize true preferences

Algorithm optimal for proposer reporting truth

Suppose:

$$w_1: f_1 \succ f_2 \succ \emptyset$$

$$w_2: f_2 \succ f_1 \succ \emptyset$$

$$f_1: w_2 \succ w_1 \succ \emptyset$$

$$f_2: w_1 \succ w_2 \succ \emptyset$$

worker:

$f_1$

$\uparrow$

$w_1$

$f_2$

$\uparrow$

$w_2$

firm:

$f_1$

$\searrow$

$w_1$

$f_2$

$\swarrow$

$w_2$

Incentives: (worker propose)

suppose  $f_2$  misreports s.t.  $f_2: w_1 \succ \emptyset \succ w_2$   
Leads to unemployed  $w_2$

$f_1$

$\uparrow$

$w_1$

$f_2$

$\uparrow$

$w_2$

$f_2$  rejects  $\Rightarrow$  unmatched

$w_2 \rightarrow f_1 \Rightarrow$  unmatched  $w_1$

$w_1 \rightarrow f_2 \Rightarrow$  firm optimal

## IC Proof by Contradiction:

- not strategy proof (at least 1 worker is incentivized to misreport)

Given true preferences,  $\succ_{w_i}$  for worker  $w_i$ , matching  $\mu$  is generated by DA when preferences submitted are  $\succ$ . There is a worker (say  $w_1$ ) that is strictly better off reporting  $\hat{\succ}_{w_1}$  (misreport). The match when  $w_1$  reports  $\hat{\succ}_{w_1}$  is  $\hat{\mu}$ .  $\hat{\mu}(w_1) \succ_{w_1} \mu(w_1)$ .

Define set  $B = \{w_i \mid \hat{\mu}(w_i) \succ_{w_i} \mu(w_i)\}$  as workers better off w/ worker 1's misreport. Define  $C = \hat{\mu}(B)$  as firms matched to  $w_i \in B$ .

- 1)  $B$  is non-empty
- 2) all  $w_i \in B$  are matched by stability  $\Rightarrow |B| = |C|$
- 3) Goal: show  $C = \mu(B)$  s.t. Set of firms are the same in  $\hat{\mu}(B)$  and  $\mu(B)$ .  $\therefore$  just re-matching where contradiction arise

Given firm  $f \in C$ , let  $w = \mu(f)$  s.t.  $w$  is original match for  $f$ . Suppose  $w \notin B$ , meaning they are matched w/ a less preferred firm. Let  $\hat{w} = \hat{\mu}(f)$  and  $\hat{f} = \hat{\mu}(w)$ .

Continuing Tuesday - Read proof in notes